

KARNAYANA ROHITH

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PROFILE

B.Tech ECE student focused on cybersecurity, Linux systems, and low-level software. Interested in offensive security, system internals, vulnerability research, and AI-assisted security tooling. Experienced with filesystem analysis, Android security research, and building autonomous cybersecurity agents.

EDUCATION

Andhra University College of Engineering <i>B.Tech in Electronics and Communication Engineering</i> Sri Gayathri Junior College, Gotlam <i>Intermediate (MPC)</i> ZPHS Siripuram <i>Secondary School Certificate (SSC)</i>	Visakhapatnam, IN 2023 – Present GPA: 8.4/10 2021 – 2023 95% 2021 85%
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TECHNICAL SKILLS

Languages: C, Python, Bash, Assembly, JavaScript

Cybersecurity: Ethical Hacking, Vulnerability Assessment and Penetration Testing (VAPT), Web Application Security, CVE Research & Exploitation, Log Analysis, IAM, OWASP Top 10, Network Reconnaissance, Digital Forensics

Networking: Computer Networking, TCP/IP, DNS, HTTP/HTTPS, Network Troubleshooting, Wi-Fi & LAN Infrastructure

Systems & OS: Linux (Arch, Ubuntu, Debian-based), Windows, macOS, Process & Memory Internals, EXT4 Filesystem Structures

AI / ML: LangGraph, Ollama, ChromaDB, PyTorch, OpenCV, CUDA, Object Detection Pipelines

Tools: Git, Docker, Linux CLI, Nmap, Wireshark, GDB, ADB, Fastboot, MTKClient

Concepts: Operating System Internals, Filesystem Structures, System Security, Hardware Attack Surfaces, Agentic AI

PROJECTS

Aegis | *Python, LangGraph, Ollama, ChromaDB, Docker* |  github.com/karnayanarohith/Aegis

- Built a dual-mode Blue Team / Red Team AI cybersecurity agent capable of autonomous offensive and defensive security tasks using LangGraph orchestration and locally-run LLMs via Ollama.
- Implemented persistent threat memory using ChromaDB vector store, enabling contextual recall of past incidents and attack patterns across sessions.
- Integrated specialized tools for log analysis, network reconnaissance, vulnerability scanning, and incident response into a unified agentic pipeline.
- Containerized the full system with Docker; designed a CLI interface for interactive and automated security workflows.
- Red Team mode generates attack simulations and adversarial payloads; Blue Team mode triages alerts, correlates logs, and recommends mitigations.

EXT4 File Recovery Tool | *C, Linux, EXT4* |  github.com/karnayanarohith/Aegis_File_Recovery_Tool

- Developed a low-level file recovery tool in C that parses raw EXT4 disk images directly, without mounting the filesystem.
- Implemented superblock and inode table parsing to identify deleted or unlinked files and reconstruct data via inode block pointers.
- Built an ncurses TUI with preview functionality, allowing inspection of recoverable file content before committing to recovery.
- Enforced read-only disk access throughout to ensure forensically sound, non-destructive recovery operations.
- Demonstrates deep understanding of EXT4 structures: block groups, inode bitmaps, data block chains, and directory entry formats.

EXPERIENCE

Network Operations Lead

March 2025

ELEXCENTRA Hackathon, Andhra University

- Sole network administrator for a 400+ participant hackathon; independently planned and deployed Wi-Fi and LAN infrastructure across six rooms.
- Configured routers and switches, ran LAN cabling, and integrated diverse client devices into a unified network under sustained high load.
- Diagnosed and resolved network downtime, latency spikes, and device overload in real time; maintained zero extended outages across the 5-hour event.

Hardware CVE Research

2025

Self-Directed Security Research, Realme C15 (MT6765)

- Conducted end-to-end MediaTek BROM exploit research on a personal Android device (RMX2180, MT6765, Android 11, July 2022 security patch).
- Successfully executed the MTKClient Download Agent authentication bypass: dumped `seccfg` partition, captured pre/post hex diff confirming `lock_state` change from `0x01` to `0x03`, and unlocked the bootloader via BROM RAM patching.
- Mapped unpatched CVE attack surface on end-of-life firmware: identified CVE-2022-20421 (Binder UAF, kernel 4.19, unpatched on device) and CVE-2024-20106 (MT6765 m4u type confusion) as high-priority targets.
- Used ADB, GDB, and `dmesg` for device profiling, baseline capture, and kernel state inspection; documented full research methodology with terminal logs and binary evidence.

ML Research Contributor

2024

NEXUS Project, under Dr. Karri Chiranjeevi

- Worked on real-time small object detection for industrial seals and tags using MMDetection, PyTorch, and CUDA.
- Benchmarked TOOD, Deformable DETR, and DINO architectures; achieved ~98% accuracy on large objects and improved small-object mAP from zero to ~0.28 via resolution scaling and data balancing.

Hackathon Participant

2024 – 2025

Smart India Hackathon; HackAP (1st Place)

- Co-built **Foundra**, a multi-agent AI startup advisor that won 1st place at HackAP; live at `foundra-main.vercel.app`.
- Participated in Smart India Hackathon and AURORA 2.0, developing solutions for satellite imagery-based illegal mining detection.

CERTIFICATIONS

Centri Security — Blue Team Junior Analyst (BTJA)

Cisco — Ethical Hacker (Module Completion)

Deloitte — Cybersecurity Job Simulation (Log Analysis)

Tata Group — Identity and Access Management Simulation

ADDITIONAL

Languages: English (Professional), Telugu (Native)

picoCTF: Active participant; building practical CTF skills in binary exploitation, forensics